Abhinav Anand

+917903960054 | abhinavkdps@gmail.com
linkedin.com/abhinav-anand-2709/ | github.com/abhikdps | abhikdps.github.io

Education

Acharya Institute of Technology Aug 2018 - Jul 2022
Bachelor of Engineering in Information Science (CGPA: 8.4 / 10.00) *Bengaluru, KA*

Experience

Red Hat India Pvt. Ltd. Oct 2024 – Present
Software Engineer *Bengaluru, KA*

- Architected and delivered TypeScript/React Backstage plugins for Red Hat Developer Hub (RHDH) integrating Ansible Automation Platform (AAP)–spanning job template execution, Execution Environment (EE) image builds with custom registry support, and self-service software template scaffolding–enabling enterprise-scale infrastructure automation for platform engineering teams.
- Designed and built catalog data ingestion pipelines with SCM source integration, progressive-loading UIs, and in-memory caching layers to support large-scale Ansible content inventories without degrading UI performance.
- Delivered full OAuth2 authentication workflows within the Backstage scaffolder for multiple SCM providers, with session state persistence across authentication redirects, encrypted secret forwarding to scaffolder actions, and pluggable backend auth modules.
- Architected a global state management layer for cross-page notification delivery and background sync-polling, decoupling asynchronous platform operations from component lifecycles to ensure reliable event propagation independent of user navigation.
- Enforced role-based access control across platform APIs and UI surfaces using the Backstage permission framework; led major framework version upgrades and maintained compatibility across concurrent product release branches with coordinated Kubernetes Helm chart deployments.
- Contributed across multiple Ansible DevTools open-source repositories–shipped Python 3.13 compatibility across ansible-lint, ansible-navigator, and ansible-creator; extended the ansible-creator CLI with new scaffold subcommands; built VS Code extension webviews for interactive scaffolding workflows; and authored E2E test suites for the Ansible RHDH portal.
- Designed and built a production **Go** Kubernetes Operator from scratch for OpenShift that automates Ansible Automation Portal deployment via a compose-via-CRD pattern, delegating infrastructure lifecycle to the RHDH Operator; authored the full OLM bundle (CSV, File-Based Catalog), implemented security hardening (SSRF prevention, RBAC scoping, NetworkPolicy, secret validation), and built end-to-end integration test infrastructure using custom **Tekton** tasks against ephemeral **OpenShift** clusters with a live Ansible Automation Platform stack via **Konflux**.

MapleLabs Solutions Pvt. Ltd. Aug 2022 – Sept 2024
Software Engineer (Contractor – Cisco) *Bengaluru, KA*

- Worked as a contracted software engineer for Cisco, contributing to the Tetration analytics platform across two successive infrastructure programs–GCP migration and Kubernetes modernization–owning Python automation, cloud infrastructure engineering, and CI/CD across the full project lifecycle.
- Engineered and automated a **40-node** Cisco Tetration cluster on Google Cloud Platform (GCP) using Python–spanning node provisioning, decommissioning, and horizontal scalability across Compute Engine, IAM, Cloud Storage, and VPC firewall rules; provisioned and version-controlled the full cluster infrastructure as code using **Terraform**.
- Architected and executed the migration of VM-based Cisco Tetration microservices to **Kubernetes**, re-platforming workloads into containerized deployments and developing Python automation to enable dynamic cluster scaling; concurrently migrated the base OS from CentOS 7 to Alma Linux 8 to improve security compliance and long-term supportability.
- Designed, configured, and maintained **Jenkins CI/CD** pipelines automating deployment, upgrades, builds, testing, and infrastructure provisioning across GCP environments; built and customized GCP machine images via shell scripting to standardize node configuration at cluster scale.
- Authored **Ansible** playbooks and roles for configuration management and orchestration across Linux-based GCP infrastructure; developed PyTest unit test suites for Python automation modules and pre-flight validation scripts to discover and verify deployment environments before production rollouts.

Projects

Adaptive Query Accelerator | *C++20, CMake, mmap* | GitHub

- Built a high-performance persistent Key-Value Store with a custom storage engine using memory-mapped I/O and slotted-page layout, achieving **1.62M writes/sec** and **10.4M scan records/sec**.
- Designed a two-tier caching architecture (L1 Record Cache + L2 Page Cache) with pluggable eviction policies (LRU, LFU, ScanResistant) and an order-1 Markov chain predictor for probabilistic prefetching (top-2 pages asynchronously) and eviction scoring, delivering **13.5x** warm vs. cold read speedup (0.11 µs vs. 1.48 µs).
- Implemented an O(1) Hash Index with reader-writer locks for thread-safe concurrent access and an append-only write strategy for maximum throughput.

Distributed Task Queue | *Python, gRPC, Apache Kafka, PostgreSQL, Redis, Kubernetes* | GitHub

- Designed and implemented a highly available microservices-based distributed task queue targeting **10M+ tasks/day** with sub-millisecond P99 latency, using Kafka (12 partitions) for task distribution and Redis for rate limiting and deduplication.
- Exposed a dual-protocol API (gRPC on port 50051 + REST on port 8080) supporting task submission, priority scheduling, retry with exponential backoff, cancellation, and real-time queue stats.
- Integrated end-to-end observability via OpenTelemetry distributed tracing, Prometheus metrics, Grafana dashboards, and Loki structured logging; deployed workers on Kubernetes with Docker Compose for local dev.

Zero-Overhead Load Balancer | *C, eBPF/XDP, libbpf, Linux Kernel* | GitHub

- Built a Layer 4 (TCP/UDP) load balancer using eBPF/XDP that routes packets at the NIC driver level, completely bypassing the Linux kernel's networking stack achieving **300K+ PPS** in generic/SKB mode with near-zero CPU overhead.
- Implemented connection affinity via a 262K-entry LRU hash map, weighted round-robin distribution, incremental L3/L4 checksum recalculation, and hitless backend reloads without detaching the XDP program.
- Designed a control plane with TCP health checks (automatic failover), real-time observability via BPF ring buffer event streaming, and per-CPU lock-free stats counters; validated with scapy-based functional tests.

Technical Skills

Languages: Python, Go, C, C++, TypeScript, JavaScript, Bash/Shell Script
Backend & Platform: Node.js, Django, Tornado, React, Backstage (RHDH), RESTful APIs, gRPC, Microservices
Cloud & DevOps: GCP, Kubernetes, OpenShift, Docker, Helm, Terraform, Ansible, Tekton, Konflux, Jenkins, GitHub Actions, CI/CD
Systems & Networking: Linux, eBPF/XDP, libbpf
Databases & Messaging: PostgreSQL, MongoDB, Redis, Apache Kafka
Observability & Testing: Prometheus, Grafana, OpenTelemetry, Loki, PyTest, Jest, Playwright, GoogleTest